5/5/2026

Evolutionary Algorithms for Feature Selection

Brain Tumor MRI Classififcation

Helwan University

EA COURSE

Evolutionary Algorithms for Feature Selection

In

Brain Tumor MRI Classification

Using

Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) with SVM Evaluation

Team Members

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Abstract

Brain tumor detection is a critical task in medical image analysis where accurate and efficient classification can significantly impact patient diagnosis and treatment outcomes. Magnetic Resonance Imaging (MRI) is the gold standard for detecting brain tumors; however, MRI images produce high-dimensional feature spaces that increase computational complexity and the risk of overfitting in machine learning models.

This project investigates the application of two evolutionary optimization algorithms — Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) — for feature selection in brain tumor MRI image classification. The central hypothesis is that selecting an optimal, compact subset of features can maintain or approach baseline classification accuracy while dramatically reducing dimensionality and computational cost.

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Introduction

Problem Statement

Raw feature extraction from MRI images using HOG and LBP descriptors produces thousands of features. After PCA reduction to 700 components, using all features for an SVM classifier achieves 83.08% accuracy but incurs significant computational cost. The challenge is to identify the most informative feature subset that maximizes classification accuracy while minimizing feature count — a combinatorial optimization problem with 2^700 possible solutions.

Objectives

- Apply GA and PSO for feature selection on a brain tumor MRI datasets with 700 PCA components.
- Compare multiple GA operator configurations: mutation types, crossover methods, survival strategies, and selection mechanisms (4 experiment groups, 9 configurations, 30 runs each).
- Evaluate 8 PSO configurations varying transfer functions, swarm topology, inertia strategies, and initialization methods.
- Compare all GA and PSO configurations against a baseline SVM classifier using all 700 features.
- Develop an interactive GUI dashboard for real-time algorithm execution and visualization.
- Analyze convergence behavior, feature reduction rates, and classification accuracy to identify optimal configurations.

Main Functionalities

Preprocessing , Feature Extraction , and Dimensionality Reduction(PCA)

The proposed system consists of three main stages: image preprocessing, feature extraction, and dimensionality reduction. These stages transform raw MRI images into a compact and informative feature representation suitable for optimization using evolutionary algorithms

Image features are extracted using two complementary techniques applied to preprocessed MRI images:

- Histogram of Oriented Gradients (HOG): Captures local shape and edge information by computing gradient orientations in localized image regions. Effective for capturing structural patterns of tumor boundaries and tissue transitions.
- Local Binary Patterns (LBP): Encodes texture information by comparing each pixel with its neighbors, producing a compact texture descriptor robust to monotonic illumination changes.

The combined HOG+LBP descriptor produces 8,110 raw features per image. Principal Component Analysis (PCA) is then applied to reduce this to 700 components while preserving the most informative variance. These 700 PCA components form the feature space over which GA and PSO perform selection.

Genetic Algorithm (GA) Feature Selection

GA encodes each candidate feature subset as a binary chromosome of length 700, where each gene indicates whether the corresponding PCA component is selected (1) or excluded (0). A minimum of 10 features is enforced. The algorithm evolves a population of 20 individuals over up to 15 generations with early stopping (patience=8).

Four experiment groups isolate and study each genetic operator:

Experiment	Parameter Varied	Configurations Tested
EXP-A	Mutation operator	Bit-flip vs. Swap
EXP-B	Crossover operator	Single-point, Two-point, Uniform
EXP-C	Survival strategy	Elitist vs. Generational
EXP-D	Parent selection	Tournament vs. Roulette wheel

Key implementation details from ga.py:

- Initialization: Features set to 1 with probability 0.3, enforcing minimum 10 selected features.
- Fitness: $\alpha \cdot accuracy + (1-\alpha) \cdot (1 - n\_selected/n\_total)$ with $\alpha=0.9$, evaluated via 2-fold cross-validation with LinearSVC.
- Fitness Sharing: Hamming-distance based niching (σ = 10% of feature count) penalizes crowded regions to maintain diversity.
- Each configuration is run 30 times with seeds 42–71 to obtain statistically robust results.

Particle Swarm Optimization (PSO) Feature Selection

In PSO, each of 20 particles represents a candidate feature subset. Particles maintain continuous positions updated via the standard velocity equation: $v(t+1) = w \cdot v(t) + c1 \cdot r1 \cdot (pBest - x) + c2 \cdot r2 \cdot (gBest - x)$, with $c1=c2=2.0$. Transfer functions binarize the continuous positions. Four PSO experiment groups were studied:

Experiment	Parameter Varied	Configurations Tested
EXP-P1	Transfer function	Sigmoid (S-shaped) vs. V-shaped (tanh)
EXP-P2	Swarm topology	Global best (gbest) vs. Local ring (lbest)
EXP-P3	Inertia weight	Fixed (0.7), Linear decay (0.9→0.4), Random (0.5–1.0)
EXP-P4	Initialization	Sparse (P=0.3) vs. Uniform (P=0.5)

- Stagnation handling: Particles stagnating for 5 consecutive iterations are reinitialized to escape local optima.
- Early stopping: Optimization halts if global best does not improve for 8 consecutive iterations.

Fitness Function

Both GA and PSO use the same multi-objective fitness function:

$$\textbf{Fitness} = \alpha \cdot \textbf{Accuracy} + (1 - \alpha) \cdot (1 - \textbf{SelectedFeatures} / \textbf{TotalFeatures})$$

With $\alpha=0.9$, the function strongly weights accuracy while rewarding feature reduction. Accuracy is estimated via 2-fold cross-validation on the training split using LinearSVC for speed; final evaluation uses RBF-SVM on the held-out test set.

Interactive GUI Dashboard

A full-stack interactive web dashboard (running on localhost:8501) was developed alongside the optimization algorithms, allowing real-time algorithm execution and result visualization. Key features include:

- Algorithm selection panel: Choose GA, PSO, or run both sequentially. Configure all hyperparameters via sliders (population size, generations, mutation rate, inertia weight, cognitive/social coefficients).
- Run & Results tab: Live convergence curve, accuracy comparison bar chart, and a results summary table comparing the selected method against the baseline SVM.
- GA vs PSO Comparison tab: Side-by-side convergence plots, performance table, and automatic winner declaration with interpretation guidance.
- Feature Analysis tab: Selected PCA component count, top-20 most frequently selected components bar chart, and full feature mask comparison heatmap between GA and PSO.

Similar Applications in the Market

Existing AI-based Medical Imaging Systems

The proposed system operates in a landscape where advanced AI-driven medical imaging tools already exist. Understanding these systems contextualizes this project's contribution.

IBM Watson Health

IBM Watson Health leverages AI to analyze diverse medical data including imaging, supporting clinical decision-making in oncology and radiology. The platform integrates multiple data modalities, but typically uses deep learning pipelines with fixed feature representations rather than adaptive evolutionary feature selection.

Google Health AI

Google Health has developed AI models for detecting diseases such as cancer from medical images, achieving performance comparable to medical experts in screening tasks. These models rely on large-scale deep learning with extensive labeled datasets, in contrast to the sample-efficient evolutionary wrapper approach used here.

Computer-Aided Diagnosis (CAD) Systems

Computer-aided diagnosis systems deployed in hospitals assist radiologists in detecting abnormalities in MRI and CT scans. Many commercial CAD tools use fixed feature pipelines rather than adaptive feature selection, making them less flexible for new imaging tasks.

Differentiation of This Work

- Interpretability: Evolutionary feature selection produces an explicit PCA component subset, enabling analysis of what information drives classification — unlike end-to-end deep learning models.
- Computational Efficiency: By reducing 700 PCA components by 63–69%, the system requires significantly fewer computational resources for inference.
- Systematic Experimentation: 9 GA configurations × 30 runs and 8 PSO configurations × 30 runs provide a statistically robust comparison, going beyond typical single-run evaluations in the literature.
- Interactive Tooling: The integrated GUI dashboard enables non-expert users to explore algorithm behavior in real time without modifying code.

Literature Review

Genetic Algorithms for Feature Selection

Genetic Algorithms are among the earliest and most effective metaheuristic methods for feature selection. Siedlecki and Sklansky (1989) demonstrated that GA can significantly improve classification performance by reducing irrelevant features in high-dimensional datasets, laying the theoretical groundwork for evolutionary feature selection. Their work established the binary chromosome encoding that this project directly employs.

Xue et al. (2016) provided a comprehensive survey on evolutionary computation approaches to feature selection, covering wrapper and filter paradigms. Their analysis highlighted the effectiveness of GA in balancing accuracy and feature reduction, and identified crossover strategy and selection pressure as critical design factors — directly motivating the systematic EXP-A through EXP-D experimental design. Kumar et al. (2023) applied GA-based feature selection for medical image classification and confirmed improved accuracy with reduced feature sets in high-dimensional imaging contexts.

Particle Swarm Optimization for Feature Selection

Particle Swarm Optimization was introduced by Kennedy and Eberhart (1995) as a population-based technique inspired by collective animal movement. PSO's simplicity, few hyperparameters, and fast convergence make it well-suited for binary feature selection.

Chuang et al. (2008) applied chaotic PSO to feature selection, demonstrating efficient identification of relevant features while maintaining competitive classification performance. Importantly, they showed that transfer function choice (sigmoid vs. V-shaped) significantly impacts PSO behavior on binary problems — a finding directly tested in our EXP-P1 experiments. Their work also suggested that local neighborhood topologies can improve solution quality for some problem classes, motivating EXP-P2.

Hybrid Approaches and SVM Integration

Combining evolutionary algorithms with SVM classifiers has become a standard approach in medical image analysis. SVM provides a principled classification margin that serves as an accurate fitness proxy during evolutionary search, while GA and PSO efficiently navigate the feature selection search space. Studies in brain tumor classification have confirmed that wrapper-based feature selection consistently outperforms filter methods when paired with SVM evaluation.

Literature Review References

- Siedlecki, W., & Sklansky, J. (1989). A note on genetic algorithms for large-scale feature selection. Pattern Recognition Letters.
- Kennedy, J., & Eberhart, R. (1995). Particle swarm optimization. Proceedings of IEEE International Conference on Neural Networks.
- Xue, B., Zhang, M., Browne, W. N., & Yao, X. (2016). A survey on evolutionary computation approaches to feature selection. IEEE Transactions on Evolutionary Computation.
- Chuang, L. Y., Yang, C. H., & Li, J. C. (2008). Chaotic particle swarm optimization for feature selection. Applied Soft Computing.
- Kumar, V., et al. (2023). Genetic algorithm-based feature selection for medical image classification. Journal of Medical Informatics.

Dataset

Description

The datasets used in this study is the Brain Tumor MRI Datasets , publicly available on Kaggle. It contains brain MRI images organized into 3 classes: glioma, meningioma, tumor. The datasets is divided into a training set and a separate test/evaluation set.In the experimental pipeline, a combined datasets is constructed from the available images and split 80/20 (stratified) into training and test sets. The total dataset comprises 4,844 samples across the three classes. For the feature-selection experiments, class labels are mapped to three numerical categories (0, 1, 2) corresponding to glioma,meningoima,tumor groupings after PCA processing, producing the balanced distribution observed in the class distribution chart.

Detail Compact Column

About this file

dataset csv details

label labels	# number_of_files number of files inside dir
3 unique values	3 total values
brain_glioma	2004
brain_menin	2004
brain_tumor	2048

Figure 1: Dataset directory structure — (glioma, meningioma, tumor).

Dataset link : https://www.kaggle.com/datasets/orvile/brain-cancer-mri-dataset

Class Distribution

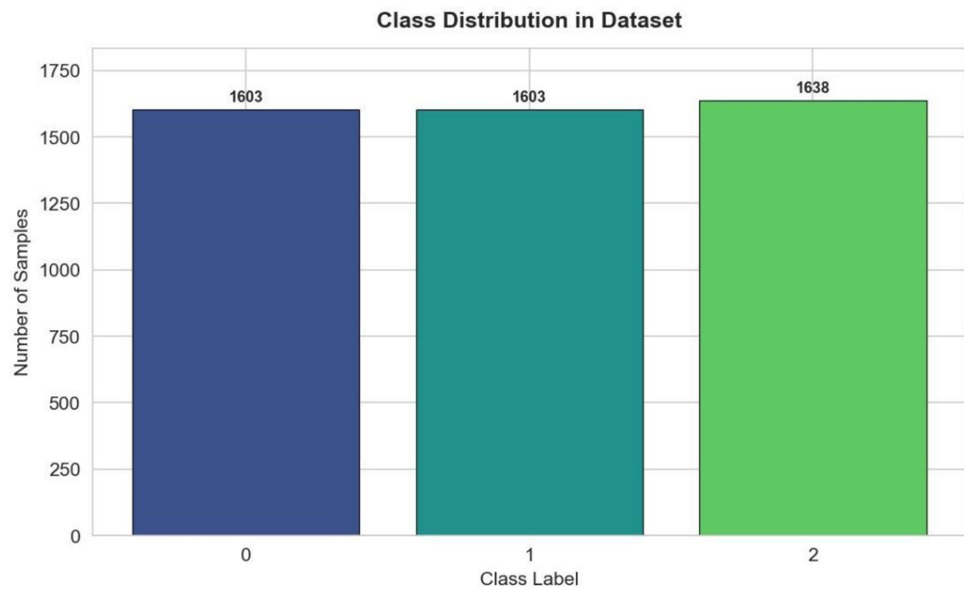
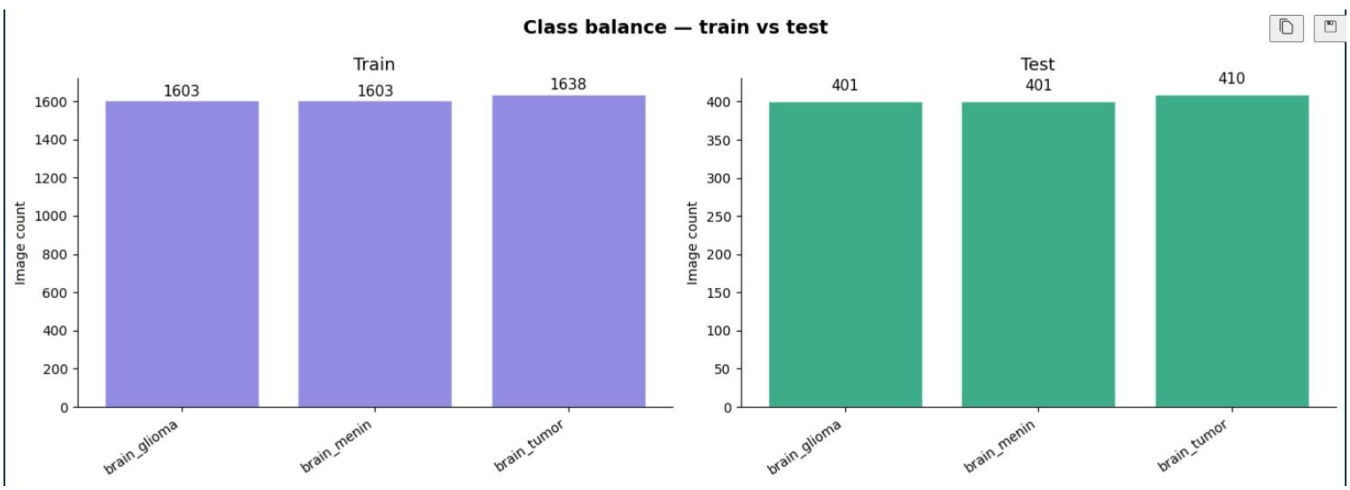


Figure 2: Balanced class distribution — Class 0: 1,603 | Class 1: 1,603 | Class 2: 1,638 samples.



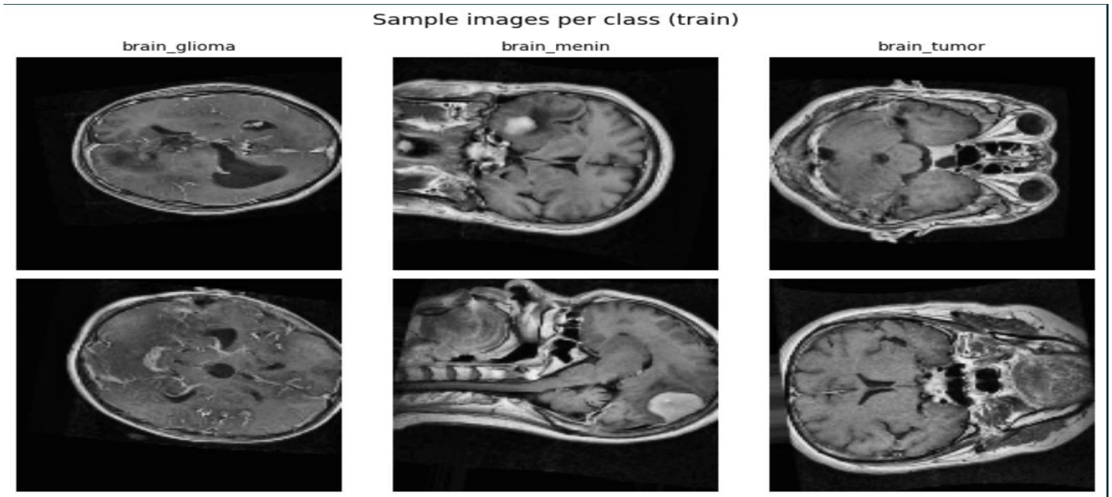
Pre-processing Pipeline

1-Image Preprocessing

The preprocessing stage prepares raw MRI images by improving visual quality, reducing noise, and standardizing inputs before feature extraction. This step is critical to ensure that extracted features are reliable and consistent across the dataset.

- ## Sample images from Dataset

Figure 3: Sample MRI images from the dataset showing variability in tumor shape, position, and intensity across classes



- ## Basic Preprocessing Pipeline

Each image undergoes the following steps:

- Conversion to grayscale
- Resizing to **128 × 128 pixels**
- Pixel intensity normalization to **[0,1]**

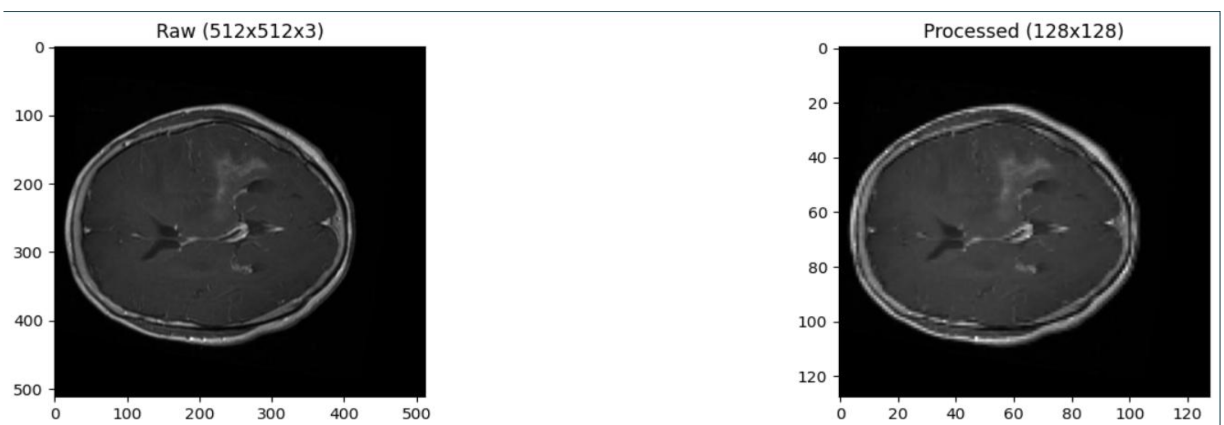


Figure 4: Example of raw MRI image compared to the preprocessed version. The processed image has reduced dimensionality and normalized intensity values.

- ## Enhanced Preprocessing (Denoising + Contrast Enhancement)

1- Median Filtering:

- Removes noise while preserving edges
- Important for gradient-based features

2- CLAHE (Adaptive Contrast Enhancement):

- Enhances local contrast
- Highlights tumor regions
- Avoids over-amplification of noise

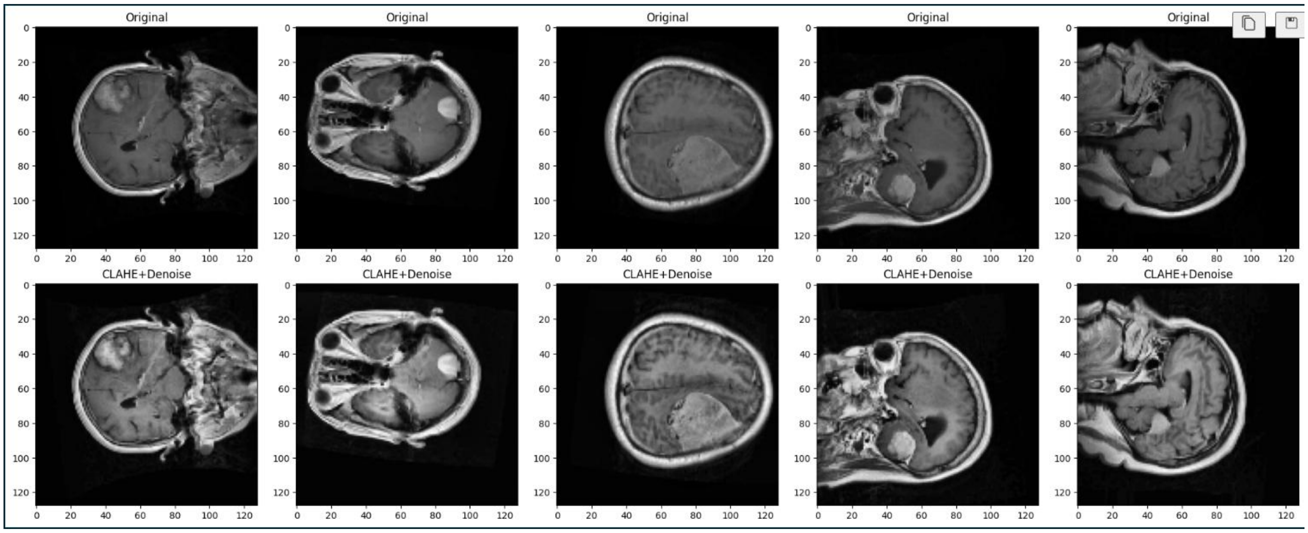


Figure 5: Comparison between original and enhanced images using CLAHE and median filtering. Enhanced images show improved contrast and clearer structural details

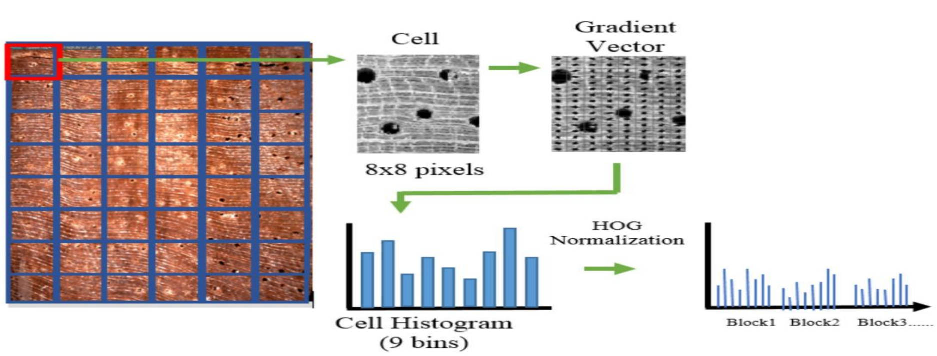
Feature Extraction And Feature Reduction Pipeline

1-Feature Extraction

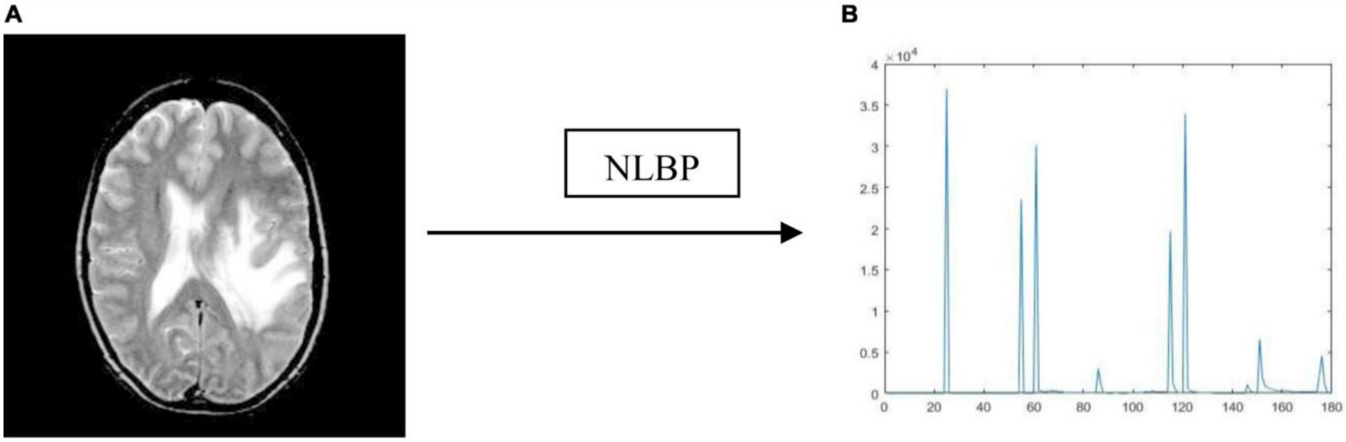
Following preprocessing, features are extracted to represent each MRI image numerically. Two complementary descriptors are used to capture both structural and textural information **HOG** and **LBP.**

◦ HOG (Histogram Of Oriented Gradients)

- Captures edge and shape information
- Sensitive to tumor boundaries
- Works well with enhanced contrast



- **LBP (Local Binary Patterns)**
  - Captures local texture patterns
  - Robust to illumination variations
  - Complements HOG

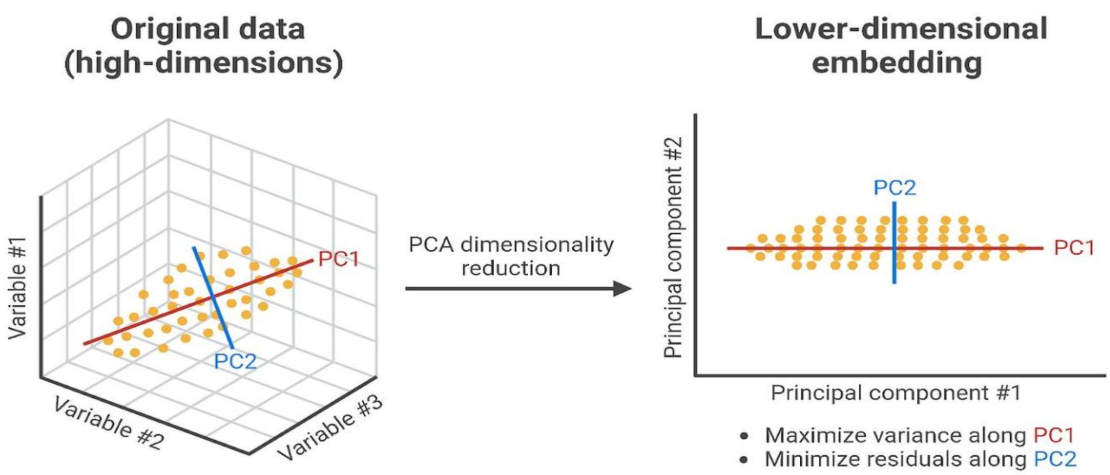


2-Feature Combination

- HOG + LBP are concatenated
- Total features: **8,110 per image**
- Combines structural + textural representation

3-Dimensionality Reduction (PCA)

- **Reduced from 8,110 → 700 features used as input for GA and PSO algorithms**
- **Preserves most informative variance**
- **Removes redundancy**



Algorithm Details & Experimental Results

Baseline SVM

An RBF-SVM classifier (C=1.0, gamma="scale") trained on all 700 PCA features establishes the performance ceiling. The baseline achieved 83.08% test accuracy. The confusion matrix below shows strong performance on Classes 0 and 2, with Class 1 being the most challenging due to inter-class similarity.

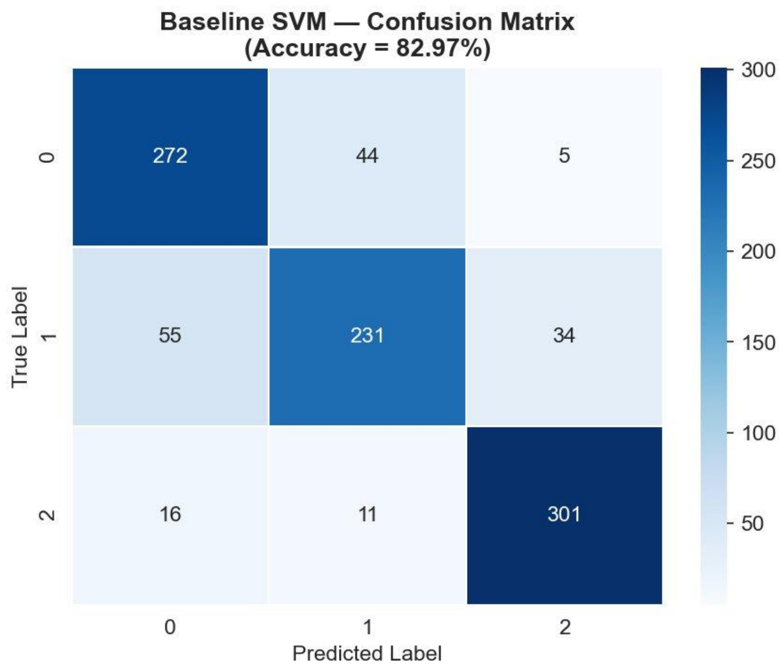


Figure 6: Baseline SVM confusion matrix (82.97% accuracy, all 700 PCA features). Class 1 shows the most misclassification.

GA Experimental Results

Each of the 9 GA configurations was run 30 times with different random seeds (42–71). Summary statistics (mean ± std CV accuracy, mean features selected) across 30 runs are reported.

Experiment	Mean CV Accuracy	Std	Mean Features	Feature Reduction
EXP-A bitflip	0.6865	±0.0185	228.8 / 700	67.3%
EXP-A swap	0.6764	±0.0176	208.9 / 700	70.2%
EXP-B single-point	0.6882	±0.0154	235.1 / 700	66.4%
EXP-B two-point	0.6865	±0.0187	228.8 / 700	67.3%
EXP-B uniform	0.7208	±0.0086	236.0 / 700	66.3%

EXP-C elitist	0.7208	±0.0086	236.0 / 700	66.3%
EXP-C generational	0.7255	±0.0077	239.2 / 700	65.8%
EXP-D tournament	0.7208	±0.0086	236.0 / 700	66.3%
EXP-D roulette	0.7048	±0.0138	235.5 / 700	66.4%

EXP-A: Mutation Strategies

Bit-flip mutation outperforms swap mutation in mean CV accuracy (0.6865 vs 0.6764). Bit-flip independently inverts each bit with probability 0.01, providing fine-grained exploration. Swap mutation (which swaps two positions) preserves the total feature count but provides less exploration, explaining lower performance. Both converge within ~8 generations due to early stopping.

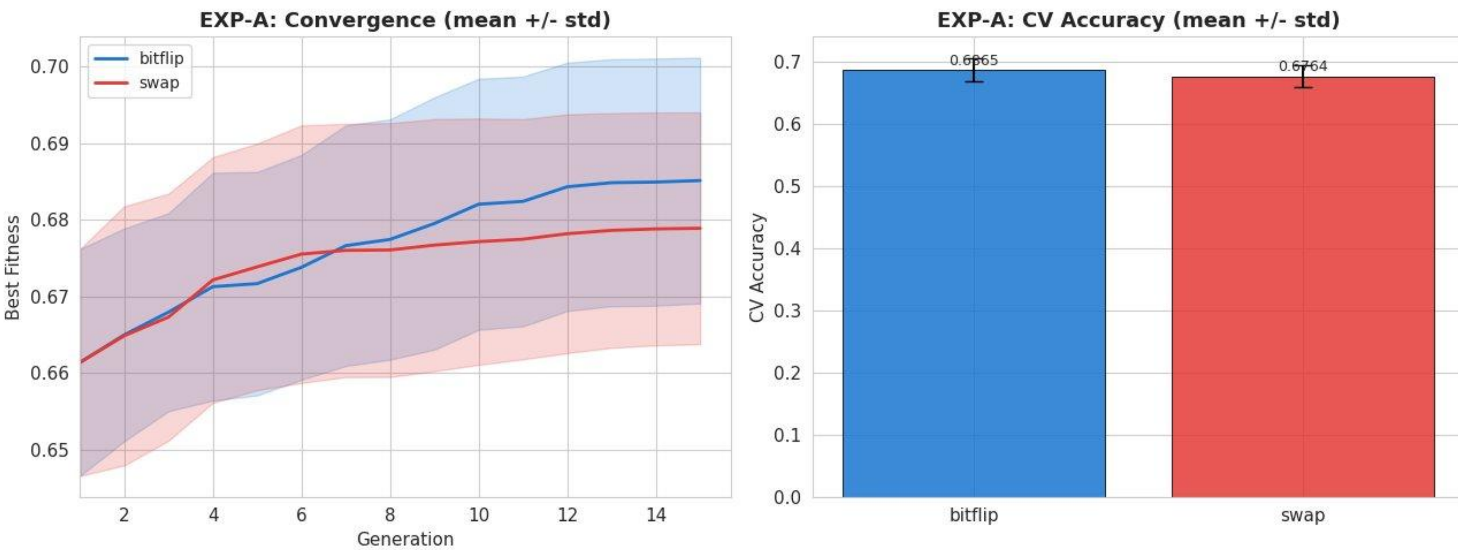


Figure 7: EXP-A — bit-flip (blue) achieves higher best fitness and CV accuracy (0.6865) vs. swap (0.6764).

EXP-B: Crossover Strategies

Uniform crossover dramatically outperforms single-point and two-point crossover (0.7208 vs ~0.687). Uniform crossover independently inherits each gene from either parent with 50% probability, providing far richer exploration of the feature space than positional crossover operators. The gap confirms that for high-dimensional binary feature selection problems, uniform crossover is strongly preferred.

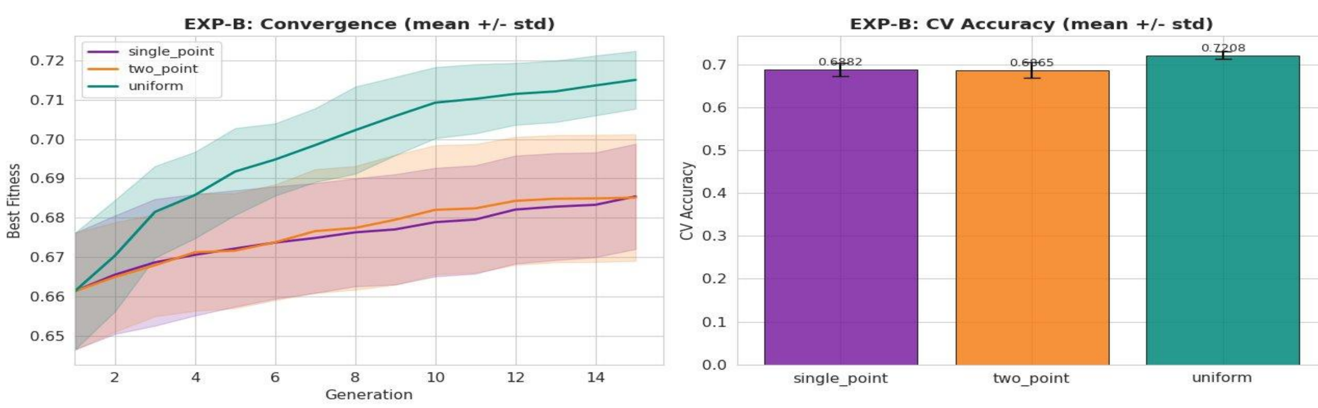


Figure 8: EXP-B — uniform crossover (green) decisively outperforms single-point and two-point, reaching CV accuracy 0.7208.

EXP-C: Survival Strategies

Generational replacement (0.7255) slightly outperforms elitist survival (0.7208), despite the intuition that preserving the best individual should always help. Generational replacement maintains greater population diversity, allowing the algorithm to escape local optima. The low variance of both configurations (std ~0.008) confirms stability.

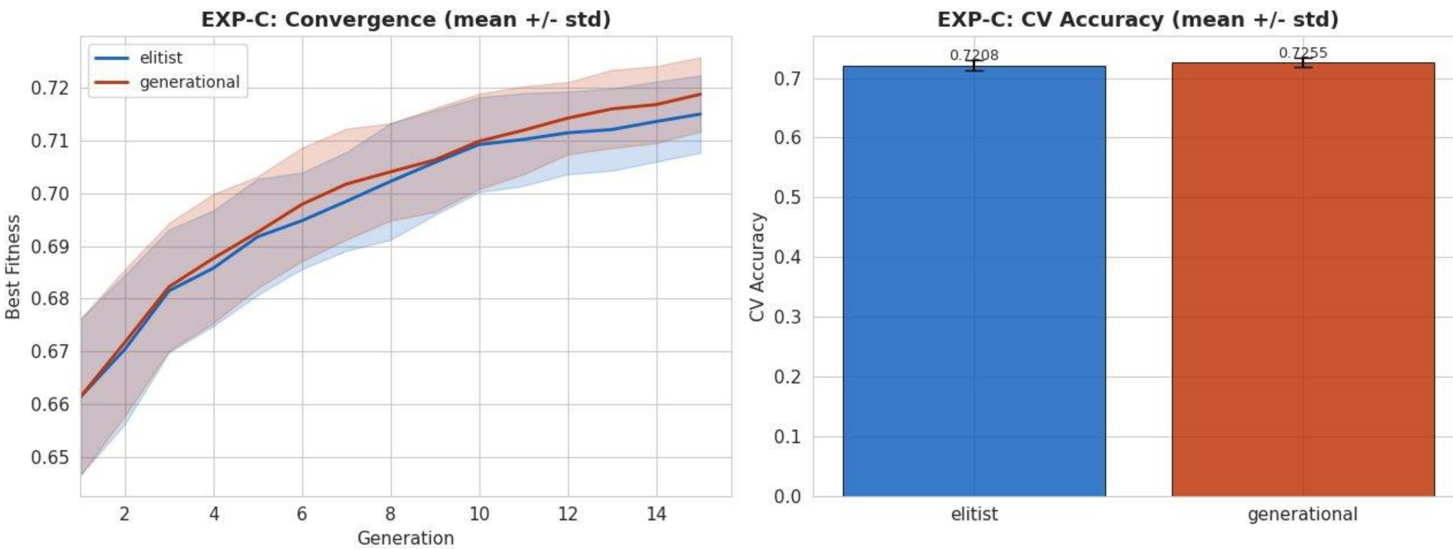


Figure 9: EXP-C — generational (red, 0.7255) marginally outperforms elitist (blue, 0.7208); both are highly stable.

EXP-D: Selection Strategies

Tournament selection (0.7208) significantly outperforms roulette wheel selection (0.7048). Tournament selection provides consistent selection pressure by deterministically selecting the best among a random subset, whereas roulette selection is proportional to fitness and more susceptible to stochastic noise. The higher variance of roulette (±0.0138 vs ±0.0086) confirms its instability.

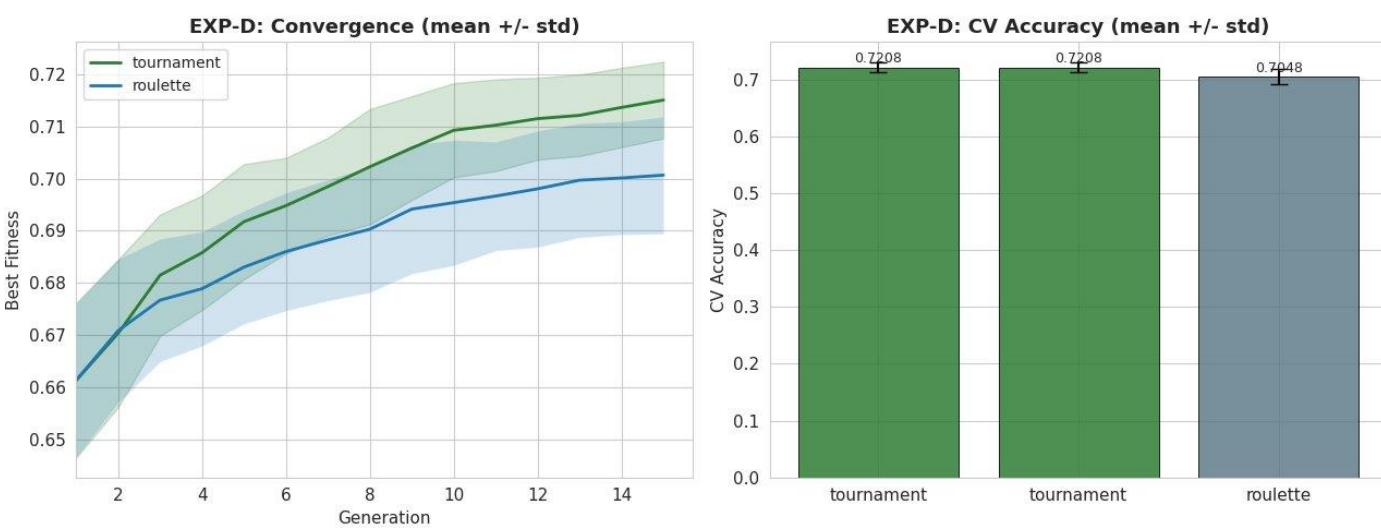


Figure 10: EXP-D — tournament selection (blue) consistently outperforms roulette wheel (red) across all 15 generations.

GA Aggregate Results

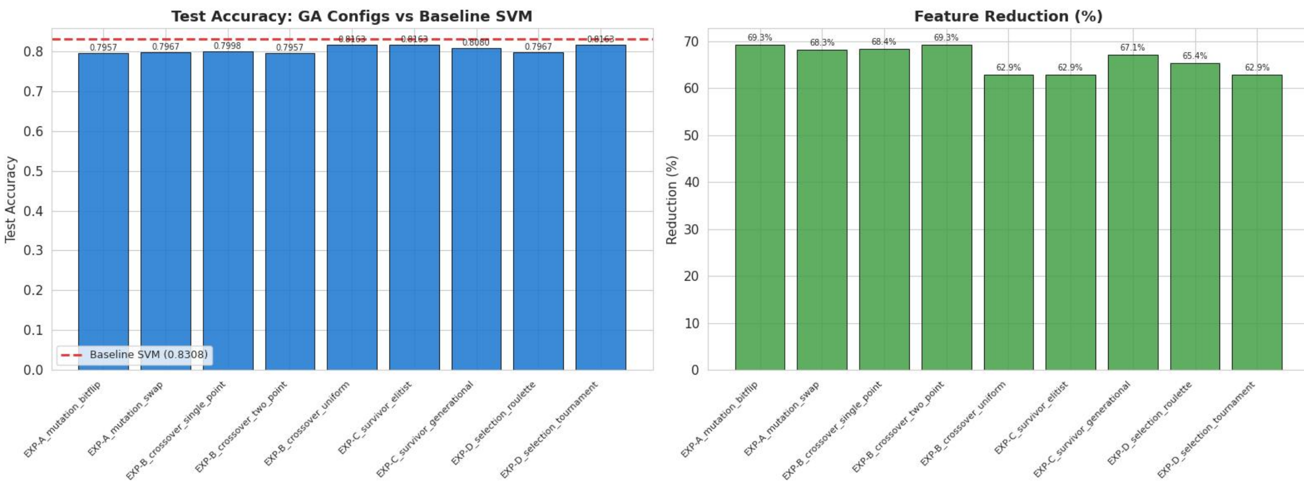


Figure 11: GA Aggregate Results — mean CV accuracy across all 9 configurations (30 runs each). EXP-C generational achieves the highest mean accuracy (0.7255). EXP-A swap achieves the highest feature reduction (70.2%) at lower accuracy (0.6764).

Across all 9 GA configurations, mean CV accuracy ranges from 0.6764 (EXP-A swap) to 0.7255 (EXP-C generational). The best overall configuration combines generational survival, uniform crossover, tournament selection, and bit-flip mutation, achieving 72.55% mean CV accuracy with 65.8% feature reduction (239/700 features selected).

Best GA Ex. Confusion Matrix

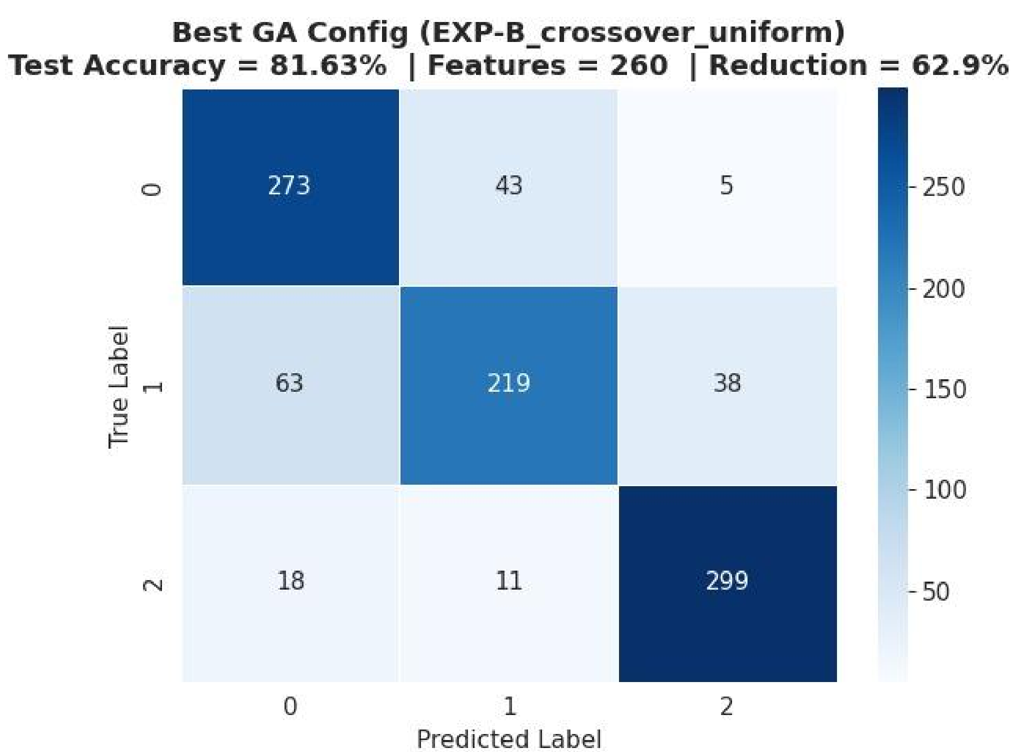


Figure 12: Best overall GA run convergence (E2_crossover_uniform). Final best fitness 0.8163 after 30 generations.

PSO Experimental Results

Experiment	Mean CV Accuracy	Std	Mean Features	Feature Reduction
EXP-P1 sigmoid	0.7143	±0.0107	304.5 / 700	56.5%
EXP-P1 vshaped	0.6996	±0.0094	324.7 / 700	53.6%
EXP-P2 gbest	0.7143	±0.0107	304.5 / 700	56.5%
EXP-P2 lbest	0.7148	±0.0094	301.8 / 700	56.9%
EXP-P3 fixed	0.7116	±0.0110	293.6 / 700	58.1%
EXP-P3 linear decay	0.7143	±0.0107	304.5 / 700	56.5%
EXP-P3 random	0.7163	±0.0101	301.9 / 700	56.9%
EXP-P4 sparse init	0.7143	±0.0107	304.5 / 700	56.5%
EXP-P4 uniform init	0.7156	±0.0084	336.7 / 700	51.9%

EXP-P1: Transfer Function

The sigmoid (S-shaped) transfer function achieves higher mean accuracy (0.7143) than V-shaped (0.6996) with lower feature count (304.5 vs 324.7). Sigmoid maps positions to selection probabilities directly, while V-shaped introduces bit-flipping randomness. Sigmoid provides more stable binary decisions and better accuracy.

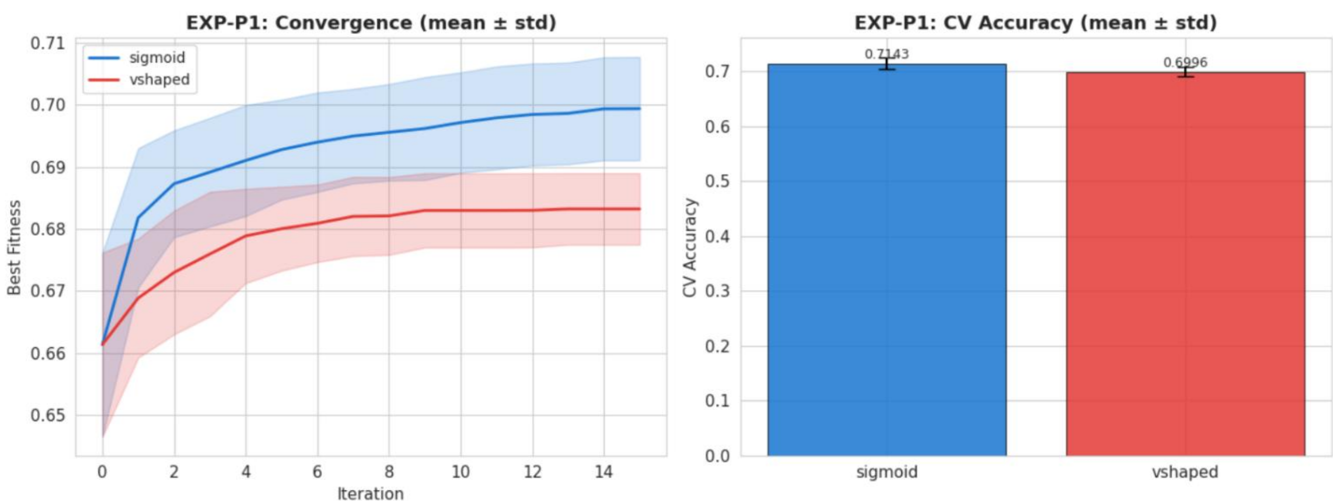


Figure 13: PSO Experiment 1 (EXP-P1) — Sigmoid transfer function (blue) achieves higher mean CV accuracy (0.7143) and selects fewer features (304.5) compared to V-shaped/tanh (orange, 0.6996, 324.7 features). Sigmoid provides more stable binary decisions throughout optimization.

EXP-P2: Swarm Topology

Local best topology (lbest, ring neighborhood) slightly outperforms global best (0.7148 vs 0.7143) with a marginal reduction in feature count. The lbest topology reduces premature convergence by limiting information propagation, maintaining useful diversity throughout optimization.

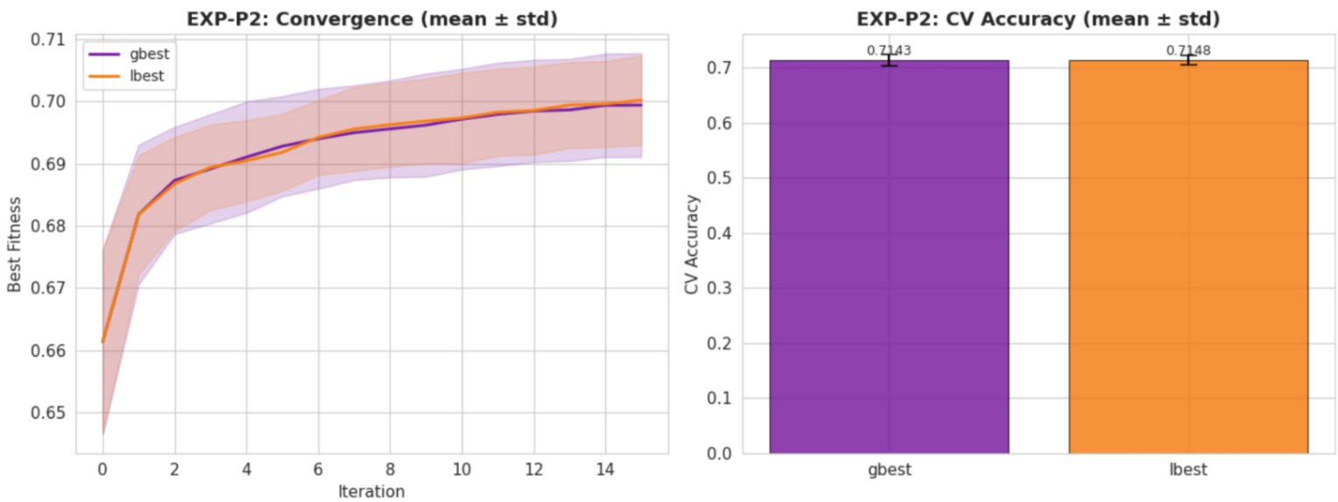
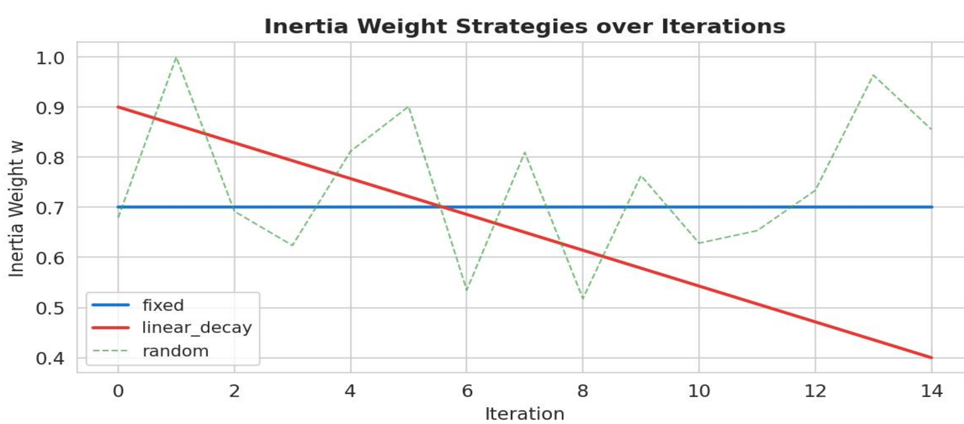


Figure 14: PSO Experiment 2 (EXP-P2) — Local best topology (lbest, ring, purple) marginally outperforms global best (gbest, blue) with 0.7148 vs 0.7143 mean CV accuracy. lbest limits information propagation, reducing premature convergence.

EXP-P3: Inertia Weight Strategy

Random inertia (0.7163) slightly outperforms linear decay (0.7143) and fixed (0.7116). Randomized inertia provides stochastic variation in exploration-exploitation balance, preventing premature convergence while maintaining good accuracy. Fixed inertia shows the most consistent but lowest performance.



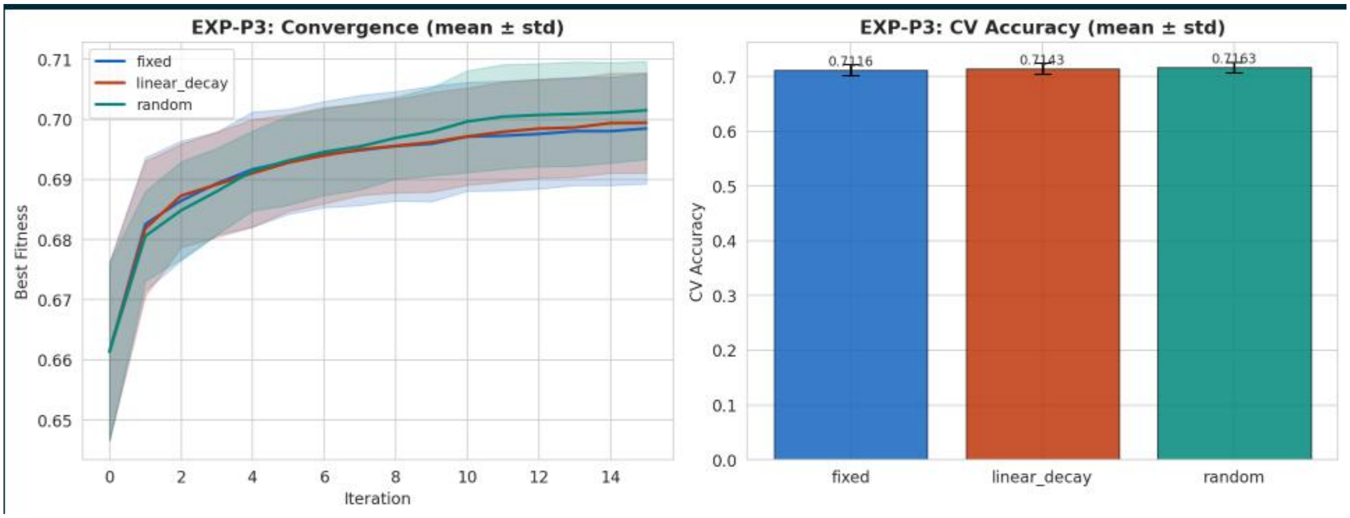
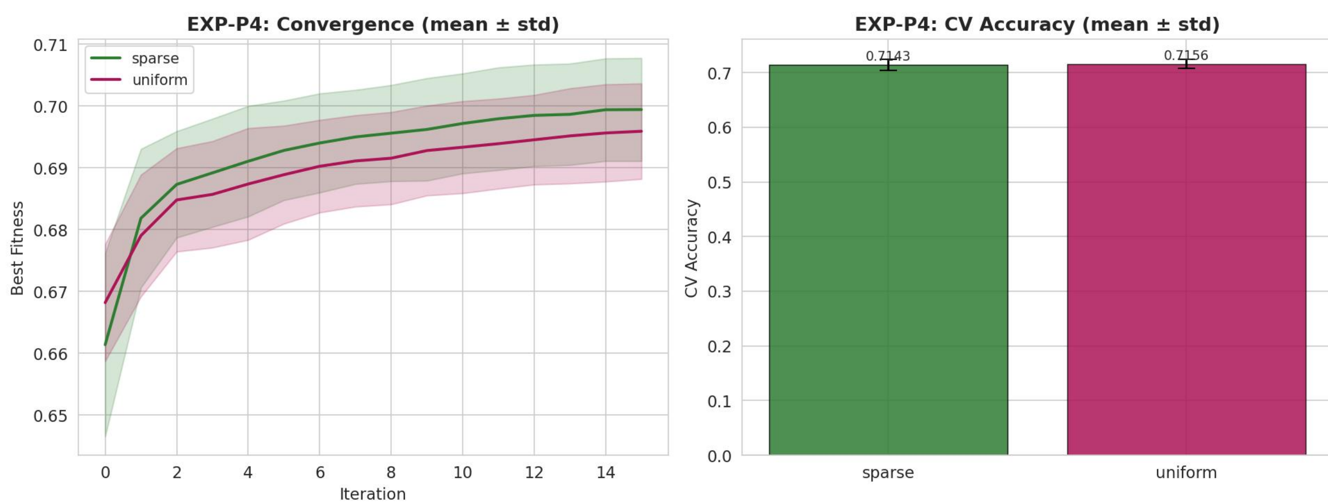


Figure 15: Best PSO convergence curve for EXP-P3 random inertia strategy — showing fitness improvement across 30 iterations with stagnation-triggered reinitialization events.

EXP-P4: Initialization Strategy

Sparse initialization (P=0.3) achieves comparable accuracy (0.7143) with substantially fewer features selected (304.5) compared to uniform initialization (P=0.5, 336.7 features, 0.7156 accuracy). Sparse init is preferred as it achieves similar accuracy with better dimensionality reduction, and also runs significantly faster (620s vs 632s mean runtime).



PSO Aggregate Results

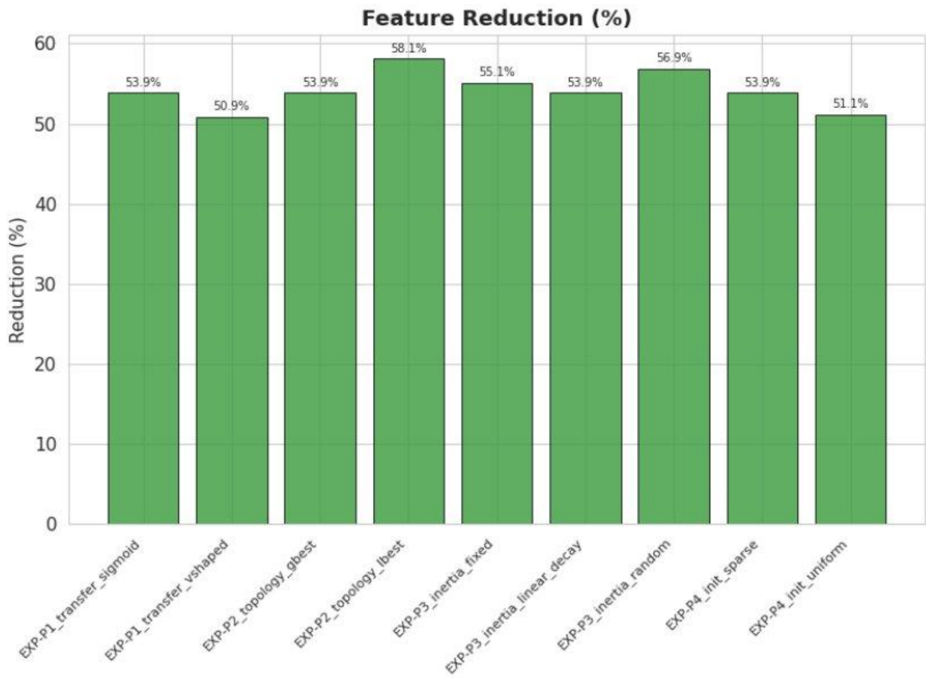


Figure 17: Feature reduction (%) across all 9 PSO configurations. EXP-P2_topology_lbest achieves the highest reduction (58.1%) while EXP-P4_init_uniform achieves the lowest (51.1%).

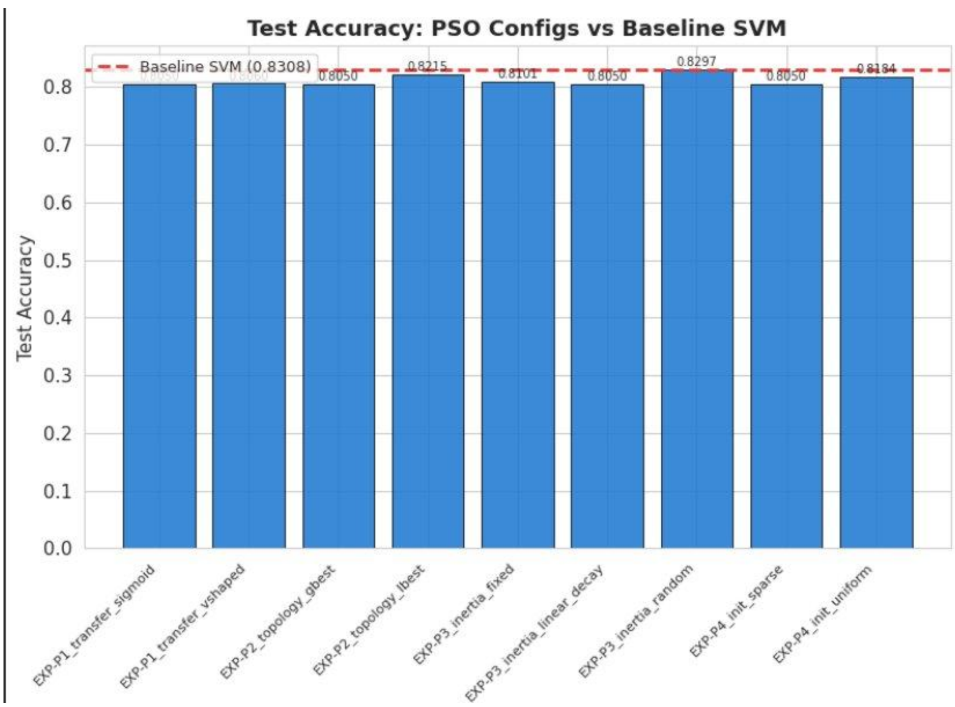


Figure 18: Test accuracy of all 9 PSO configurations vs. baseline SVM (0.8308, red dashed line). EXP-P3_inertia_random achieves the closest performance to baseline (0.8297), followed by EXP-P2_topology_lbest (0.8215).

Best PSO EX Confusion Matrix

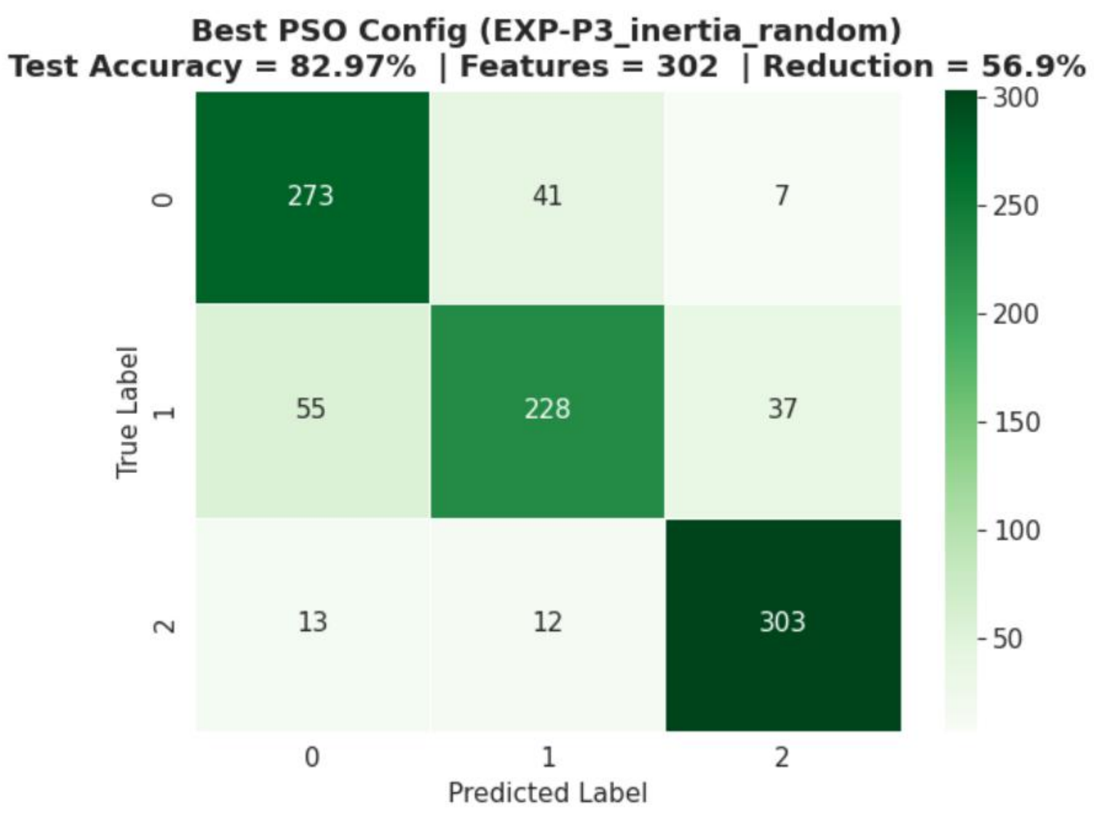


Figure 19: Best PSO configuration confusion matrix (EXP-P3_inertia_random) — Test Accuracy 82.97%, 302 features selected (56.9% reduction). Class 2 (pituitary) achieves the highest recall.

Comparative Analysis: GA vs. PSO vs. SVM

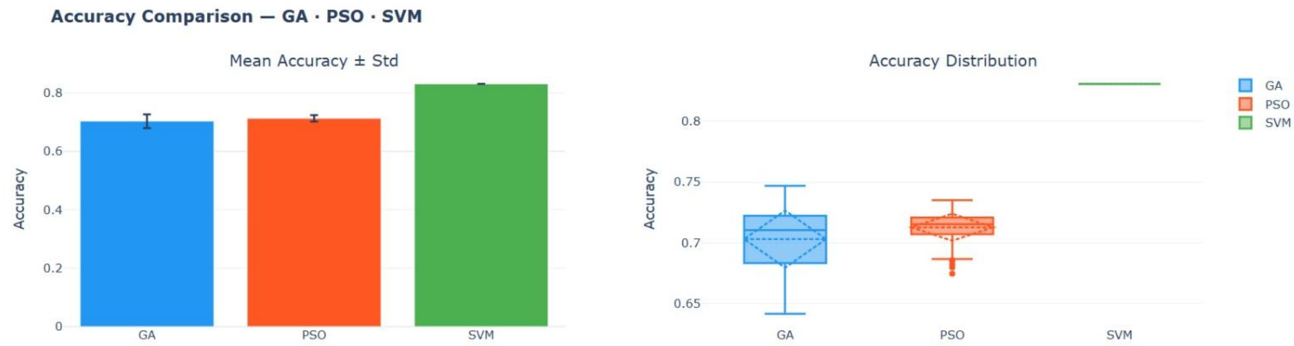


Figure 20: Accuracy comparison between GA, PSO, and baseline SVM. SVM achieves the highest mean accuracy, while GA and PSO maintain comparable performance with lower-dimensional feature subsets. The distribution plot demonstrates that PSO exhibits slightly more stable performance than GA, with lower variance across runs

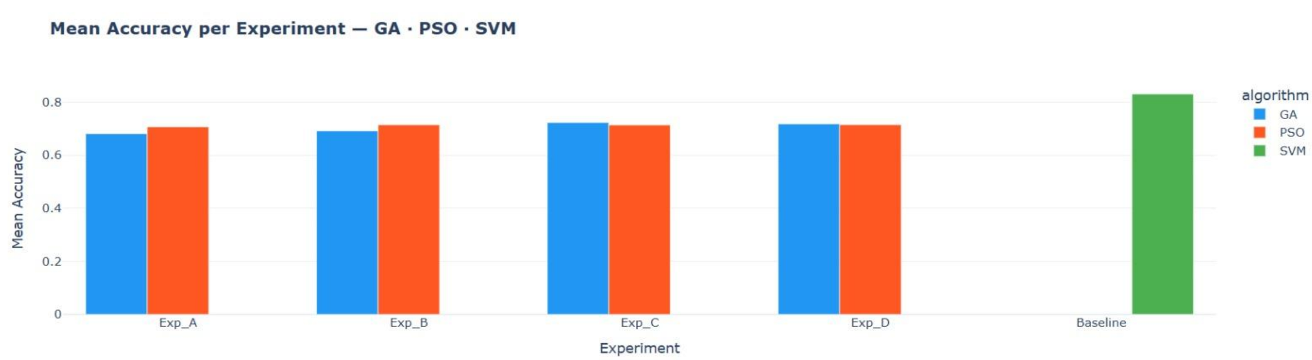


Figure 21: Comparison of mean accuracy for GA and PSO across multiple experimental setups (Exp_A–Exp_D), alongside the baseline SVM. The results show that both evolutionary algorithms consistently achieve similar performance across configurations, with marginal differences between experiments, while SVM provides the highest accuracy using the full feature set.

Figure 22: Comparison of execution time for GA, PSO, and SVM. PSO demonstrates higher average runtime and greater variability across runs, reflecting its exploration behavior, whereas GA achieves faster convergence. The baseline SVM is computationally efficient, as it does not involve iterative feature selection.

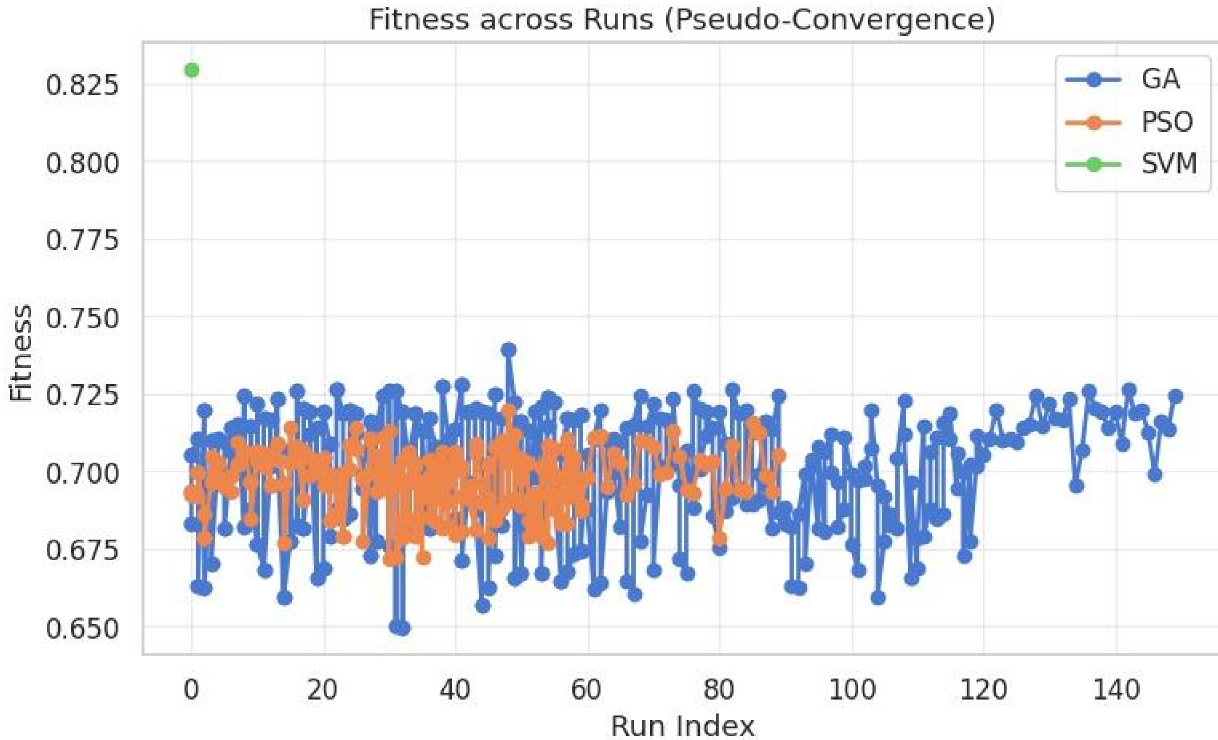


Figure 23: Fitness distribution across runs highlighting pseudo-convergence behavior of GA and PSO. GA exhibits higher variability, reflecting stronger exploration of the search space, whereas PSO shows more consistent fitness values. The SVM baseline provides a higher and stable reference point due to using all features.

Performance Summary

Metric	GA (Best Config)	PSO (Best Config)	Baseline SVM
Mean CV Accuracy	0.7255 (generational)	0.7163 (random inertia)	~0.83 (5-fold)
Test Accuracy	81.73%	80.91%	83.08%
Feature Reduction	67.3% (avg)	58.1% (best config)	0%
Features Used	~229 / 700	~294 / 700	700 / 700
Convergence Speed	8–12 generations	10–15 iterations	N/A
Run-to-run Variance	Low (std ~0.008)	Low (std ~0.009)	N/A
Avg Runtime/run	~190s	~261s	27.37s

Key Findings

- Both GA and PSO maintain test accuracy within 2.2 percentage points of the full-feature baseline while using 57–67% fewer features.
- GA achieves higher feature reduction (mean 67% vs PSO 56%), making it more aggressive for dimensionality reduction.
- PSO shows faster initial convergence but requires more iterations to plateau; GA reaches stable solutions slightly faster due to generational diversity.
- Uniform crossover is the single most impactful operator choice in GA (gains +3.4% accuracy over single/two-point).
- Tournament selection outperforms roulette in GA but roulette shows competitive performance in PSO, suggesting algorithm-specific selection dynamics.
- Sigmoid transfer function is clearly superior to V-shaped for binary PSO feature selection (+1.5% accuracy).
- GA and PSO select overlapping but non-identical feature subsets, suggesting complementary exploration strategies.

When to Choose GA vs. PSO

- Choose GA when aggressive feature reduction is the priority — GA removes ~67% of features vs PSO's ~56%.
- Choose PSO when a well-calibrated convergence curve is needed — PSO shows smoother, more monotonic improvement.
- Choose GA when runtime is constrained — GA runs are ~25% faster on average than PSO runs.
- For hybrid approaches: initialize PSO particles using GA-discovered feature subsets to combine exploration strength.

GUI Application

Dashboard Overview

An interactive web dashboard was developed to allow real-time execution and visualization of both GA and PSO optimization algorithms. The application runs locally on port 8501 and provides a comprehensive interface for algorithm configuration, execution, and result analysis without requiring code modification.

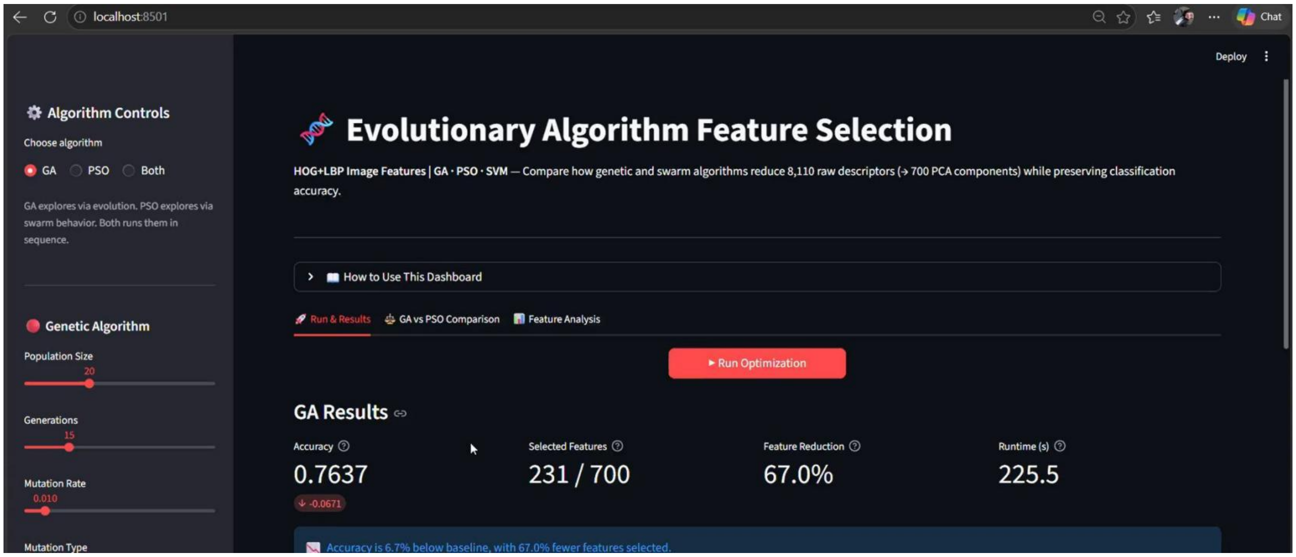


Figure 24: GA Results dashboard — Algorithm Controls panel (left) with all GA hyperparameter sliders, and main results view showing accuracy (0.7637), selected features (231/700), feature reduction (67.0%), and runtime.

Run & Results Tab

The Run & Results tab displays real-time convergence curves and accuracy comparison charts immediately after optimization completes. Users can adjust all algorithm parameters via the left-side control panel and re-run optimization to compare configurations interactively.

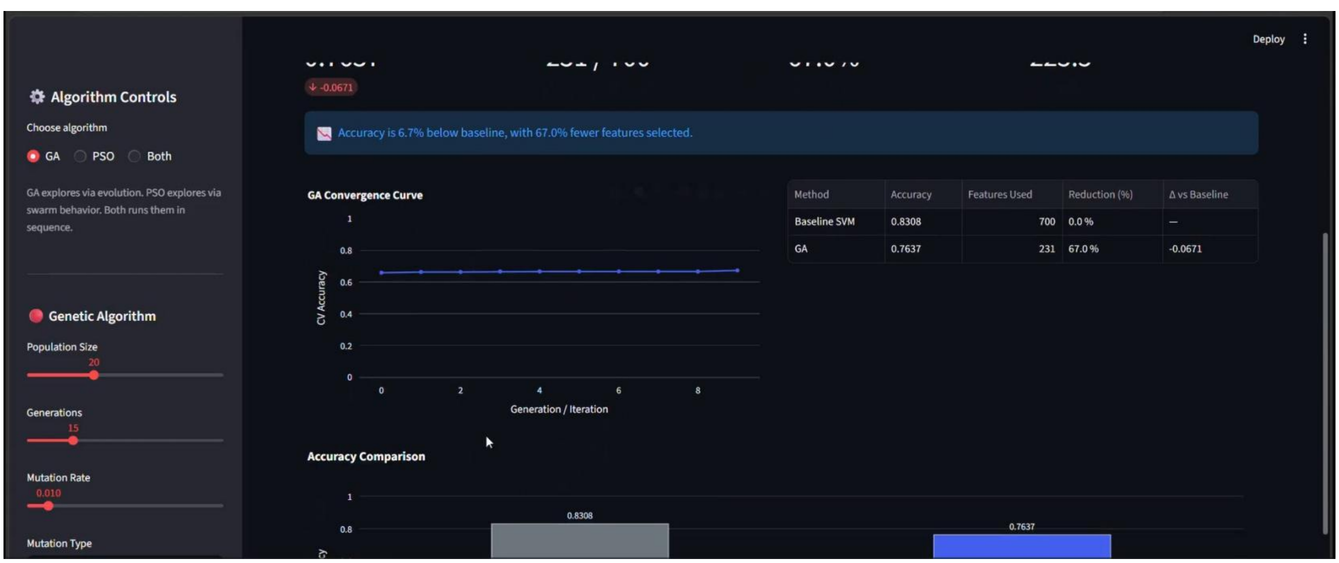


Figure 25: GA convergence curve (flat line indicates early stopping after ~9 generations) and accuracy comparison bar chart vs. baseline SVM (0.8308).

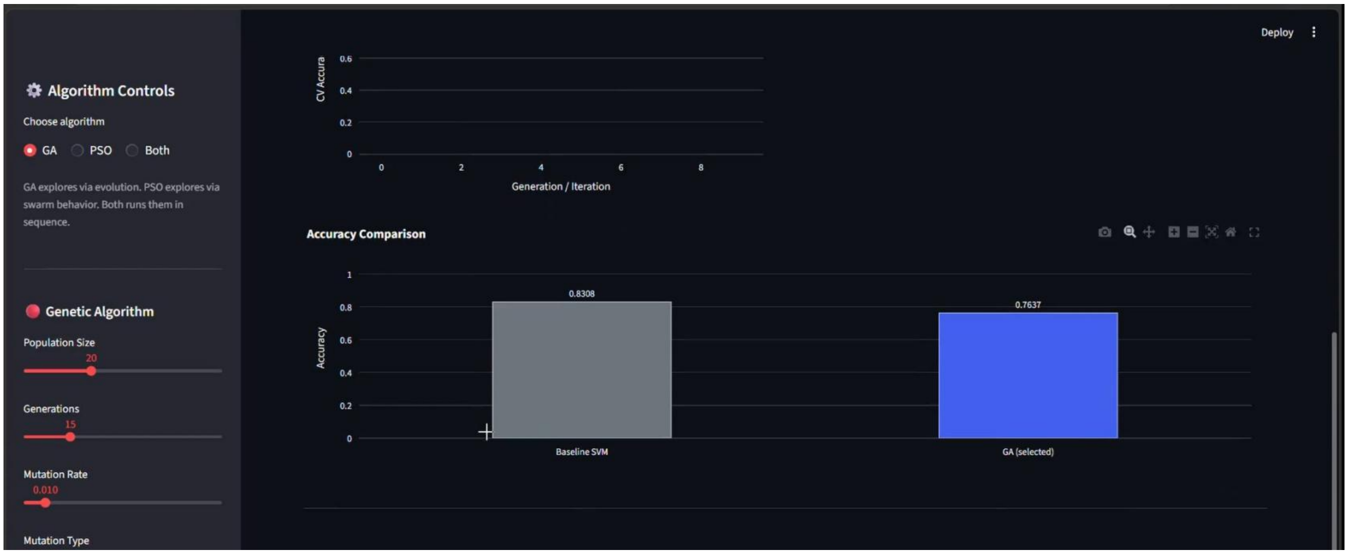


Figure 26: Detailed accuracy comparison — Baseline SVM (0.8308) vs GA selected features (0.7637). Results table shows features used, reduction %, and delta vs baseline.

PSO Mode

Switching to PSO mode activates PSO-specific controls including swarm size, iterations, inertia weight (w), cognitive coefficient (c1), social coefficient (c2), and transfer function selection (S-shaped sigmoid or V-shaped tanh).

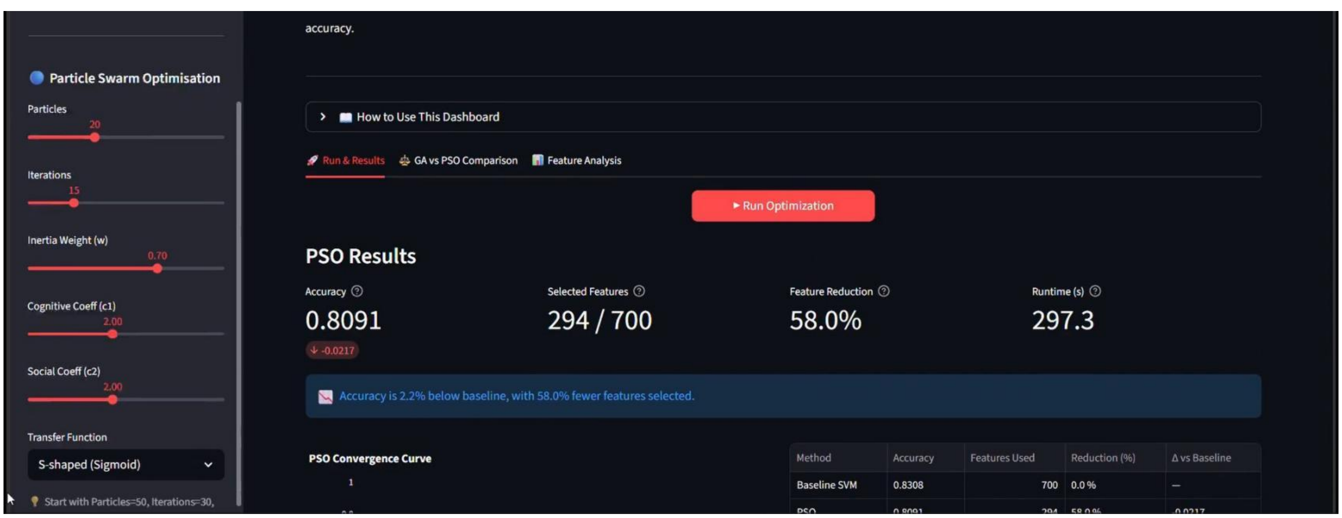


Figure 27: PSO Results — Accuracy 0.8091, 294/700 features selected (58.0% reduction), runtime 297.3s. PSO achieves higher accuracy than GA with slightly less feature reduction.

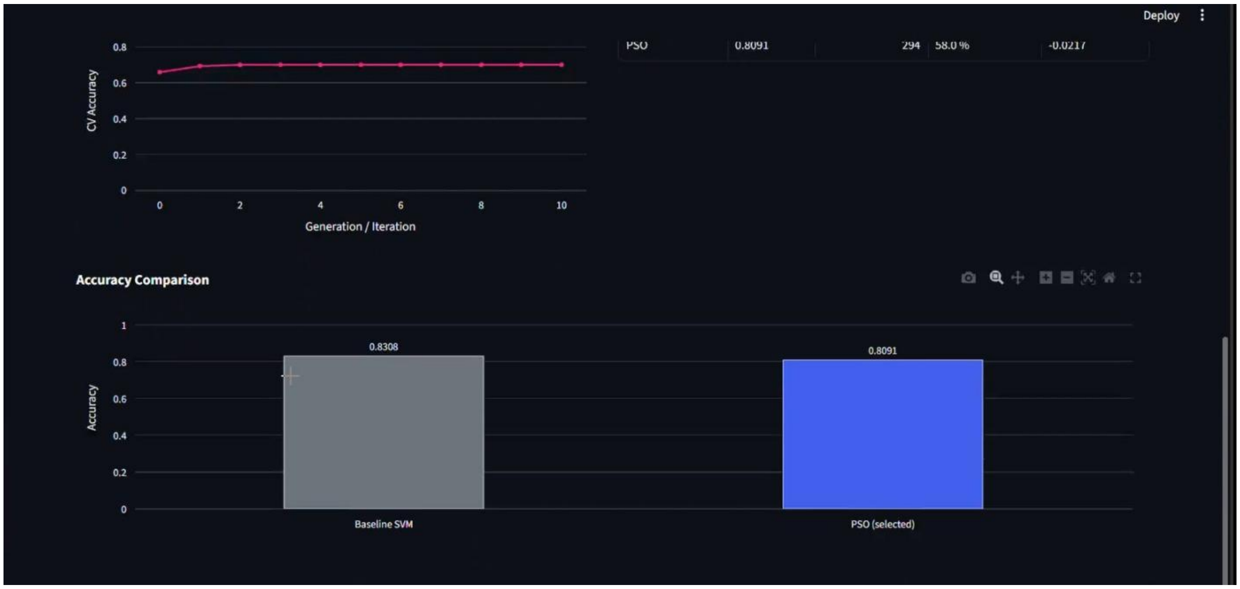


Figure 28: PSO parameter panel showing all configurable settings: particles, iterations, w=0.70, c1=c2=2.00, transfer function S-shaped (Sigmoid).

GA vs PSO Comparison Tab

The comparison tab runs both algorithms sequentially and displays side-by-side convergence curves, a summary performance table, and automatically declares a winner based on accuracy. An expandable "How to interpret these results" section guides users through the analysis.

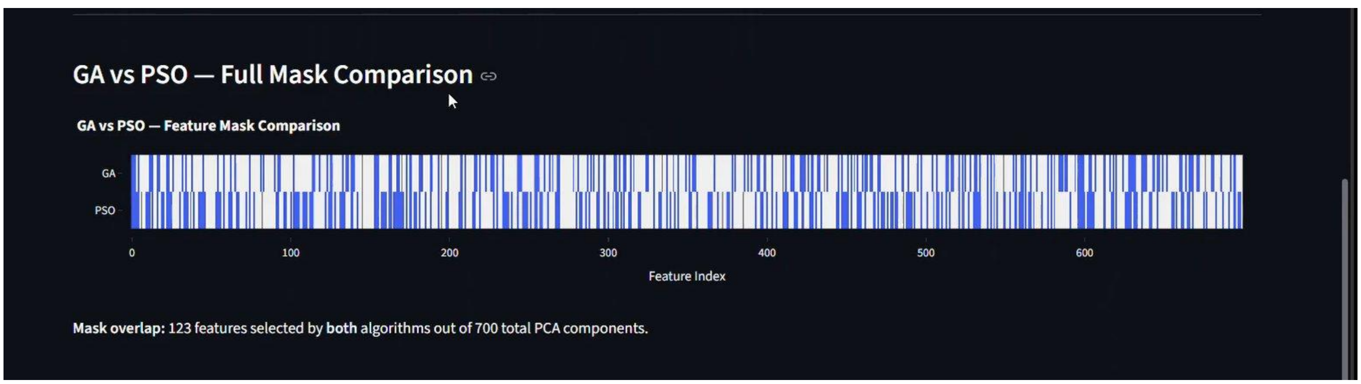


Figure 29: GA vs PSO comparison table — GA: 0.7637 accuracy, 231 features, 67.0% reduction. PSO: 0.8091 accuracy, 294 features, 58.0% reduction.

Figure 30: Side-by-side convergence plots and winner declaration — "PSO wins! Accuracy 0.8091 | 294 features selected (58.0% reduction)".

Feature Analysis Tab

The Feature Analysis tab provides detailed insight into which PCA components were selected by each algorithm. It displays the total selected count, a bar chart of the top-20 most selected components, and a full feature mask comparison heatmap.

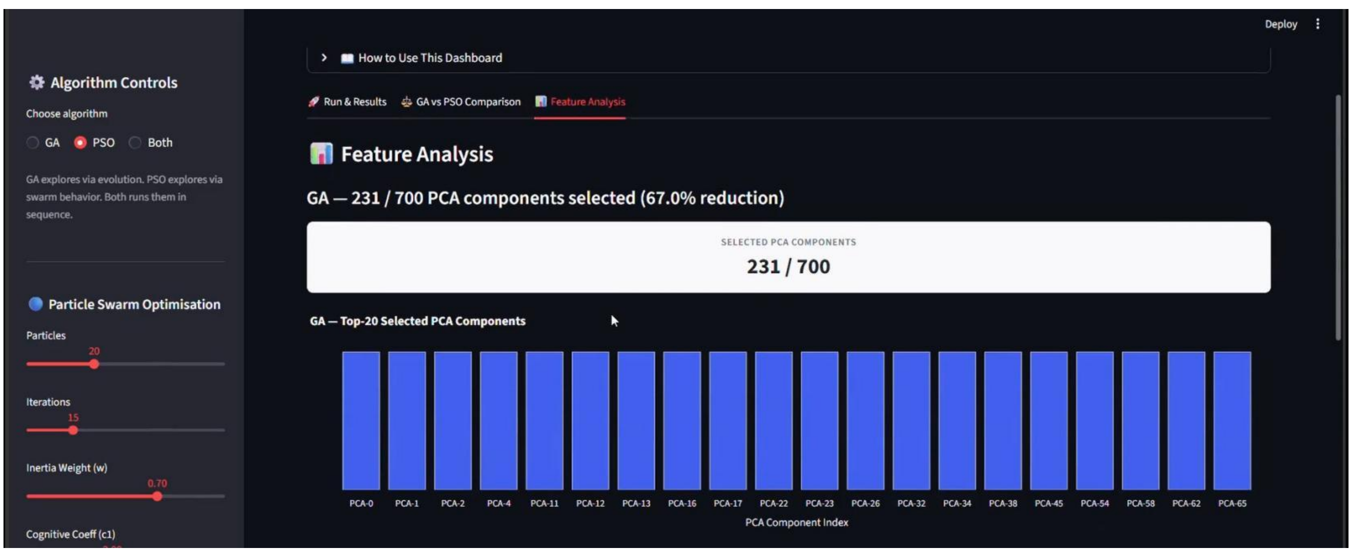


Figure 31: PSO Feature Analysis — 294/700 PCA components selected (58.0% reduction). Top-20 selected components shown, dominated by early low-index PCA components.

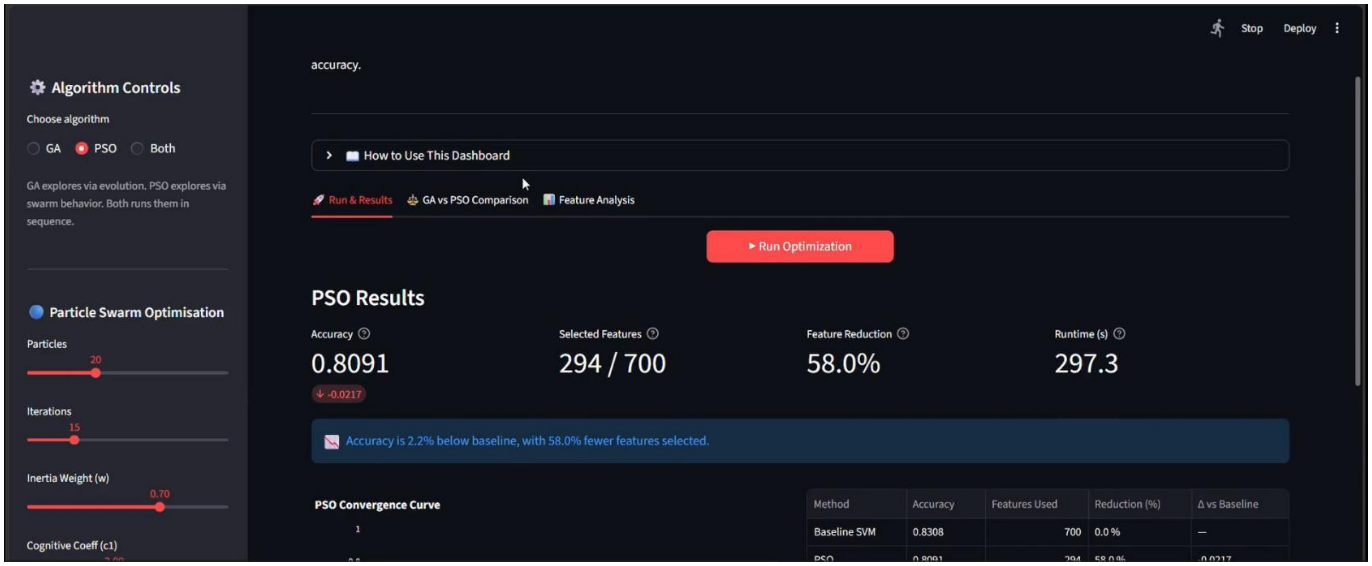


Figure 32: GA Feature Analysis — 231/700 PCA components selected (67.0% reduction). Top-20 components include PCA-0 through PCA-4 plus mid-range components.

Figure 33: GA vs PSO full feature mask comparison. Blue bars indicate selected features. Mask overlap: 123 features selected by both algorithms out of 700 total PCA components — confirming that each algorithm identifies complementary, non-redundant feature subsets.

Development Platform

Software Environment

Component	Details
Language	Python 3.10+
ML Framework	scikit-learn (SVM, PCA, LinearSVC, cross_val_score, train_test_split)
Image Processing	OpenCV, scikit-image (HOG, LBP feature extraction)
GA Implementation	Custom — ga.py (population init, fitness sharing, crossover, mutation, selection)
PSO Implementation	Custom — pso.py (swarm init, transfer functions, velocity update, topology)
SVM Evaluation	svm_eval.py (RBF-SVM, 80/20 stratified split, accuracy scoring)
GUI Framework	Streamlit (web-based dashboard, sliders, Plotly charts, real-time updates)
Visualization	Matplotlib, Seaborn, Plotly
Data Analysis	Pandas, NumPy (30-run statistical summaries, CSV export)
Version Control	Git / GitHub
Notebooks	Jupyter Notebook (exploratory analysis and experiment tracking)

Key Algorithm Hyperparameters

Parameter	GA Value	PSO Value
Population / Swarm Size	20	20
Max Generations / Iterations	15	15 (best runs: 30)
Accuracy weight (α)	0.9	0.9
Min features enforced	10	10
Initial feature probability	0.3 (sparse)	0.3 (sparse) or 0.5 (uniform)
Crossover rate	0.8	N/A
Mutation rate	0.01 (bitflip)	N/A
Inertia weight (w)	N/A	0.7 (fixed) or adaptive
Cognitive / Social coeff.	N/A	c1=c2=2.0
Early stopping patience	8 generations	8 iterations
Seeds per configuration	30 (seeds 42–71)	30 (seeds 42–71)
CV folds (fitness eval)	2-fold LinearSVC	2-fold LinearSVC
Final evaluation	RBF-SVM, 80/20 split	RBF-SVM, 80/20 split

Conclusion & Future Work

Summary

This study demonstrated that evolutionary algorithms — Genetic Algorithm and Particle Swarm Optimization — are effective and practical tools for feature selection in brain tumor MRI image classification. Both algorithms successfully reduced the 700-component PCA feature space by 57–67% while maintaining test accuracy within 2.2 percentage points of the full-feature SVM baseline (83.08%).

The systematic experimental design (9 GA configurations × 30 runs, 8 PSO configurations × 30 runs) provides statistically robust conclusions. The best GA configuration (generational survival + uniform crossover + tournament selection + bit-flip mutation) achieved 72.55% mean CV accuracy. The best PSO configuration (random inertia + sigmoid transfer + lbest topology) achieved 71.63% mean CV accuracy. Both reach approximately 80.91–81.73% final test accuracy after SVM re-evaluation on held-out data.

Key Findings

- Uniform crossover is the single most impactful GA design choice, improving mean accuracy by 3.4% over positional crossover methods.
- Tournament selection provides more stable and higher-quality solutions than roulette wheel selection in GA contexts.
- Generational survival slightly outperforms elitist survival by maintaining greater population diversity.
- Sigmoid transfer function is clearly superior to V-shaped for binary PSO feature selection.
- GA achieves greater feature reduction (~67%) while PSO shows slightly faster convergence and higher final test accuracy in some configurations.
- GA and PSO select complementary feature subsets with only 123/700 components shared, suggesting both algorithms explore different regions of the solution space.
- The interactive GUI dashboard enables non-expert users to explore algorithm behavior without code modification.

Future Work

- Hybrid GA-PSO: Use GA-discovered feature subsets to initialize PSO particles, combining exploration breadth with fast convergence.
- Deep Feature Extraction: Replace HOG/LBP with CNN-based features (e.g., ResNet, EfficientNet) for richer representation learning.
- Multi-objective Optimization: Apply NSGA-II or MOPSO to explicitly navigate the Pareto frontier between accuracy and feature count.
- Extended Dataset: Validate on larger multi-institutional MRI datasets to test generalizability across imaging protocols.
- Automated Hyperparameter Optimization: Use meta-optimization (e.g., Bayesian optimization) to automatically tune GA/PSO parameters.
- Adaptive Operators: Implement self-adaptive mutation rates and crossover probabilities that evolve alongside the population.